
MECHANICS : Momentum, Impulse, Collision

Linear Momentum

Definition

The **linear momentum** (or simply *momentum*) of a particle is a vector quantity defined by

$$\vec{p} = m \vec{v},$$

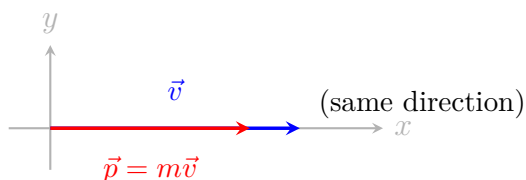
where m is the mass of the particle and $\vec{v}$ is its velocity vector.

Remarks:

- Mass m is a positive scalar, hence $\vec{p}$ and $\vec{v}$ point in the same direction.
- The SI unit of linear momentum is

$$\text{kg} \cdot \text{m s}^{-1} \quad (\text{commonly written as } \text{kg m s}^{-1}).$$

Visual Representation: Direction of Momentum and Velocity



Both vectors lie along the same line, indicating that $\vec{p}$ and $\vec{v}$ share the same direction since $m > 0$.

Impulse

Basic Idea

The momentum of a body cannot change unless a net external force acts on it.

If a force acts on a body for a short time (like when a bat hits a ball), it can produce a large change in momentum. This short-lived but strong force is called an **impulsive force**, and the effect of that force is called an **impulse**.

When a time-dependent force $\vec{F}(t)$ acts on an object, Newton's second law states:

$$\vec{F} = \frac{d\vec{p}}{dt}.$$

Rewriting this gives:

$$d\vec{p} = \vec{F}(t) dt.$$

If we add up (integrate) all the small changes during the time of contact — from t_i to t_f — we get:

$$\int_{t_i}^{t_f} d\vec{p} = \int_{t_i}^{t_f} \vec{F}(t) dt.$$

The left-hand side is the total change in momentum:

$$\Delta\vec{p} = \vec{p}_f - \vec{p}_i.$$

The right-hand side defines the **impulse** $\vec{J}$:

$$\vec{J} = \int_{t_i}^{t_f} \vec{F}(t) dt.$$

Impulse–Momentum Theorem:

$$\boxed{\Delta\vec{p} = \vec{J}}$$

It means the change in an object's momentum equals the impulse of the force acting on it.

In simple words:

- A large force acting for a short time can cause a big change in momentum.
- Impulse depends on both the strength of the force and how long it acts.
- If the force is constant, impulse can be written as $\vec{J} = \vec{F} \Delta t$.

Problem 1

Problem 1: Impulse and Average Force

A 1.2 kg ball drops vertically onto a floor, hitting it with a speed of 25 m/s. It rebounds with a speed of 10 m/s.

Questions: (a) What impulse acts on the ball during the contact? (b) If the ball is in contact with the floor for 0.020 s, what is the magnitude of the average force on the floor from the ball?

Solution 1

Impulse:

Impulse is the change in momentum:

$$J = \Delta p = p_{\text{final}} - p_{\text{initial}}$$

Take downward as positive. The ball initially moves downward at $v_i = 25 \text{ m/s}$ and rebounds upward at $v_f = -10 \text{ m/s}$. Mass $m = 1.2 \text{ kg}$.

$$J = m(v_f - v_i) = 1.2(-10 - 25) = 1.2(-35) = -42 \text{ kg} \cdot \text{m/s}$$

The negative sign indicates the impulse acts upward.

Average force:

Impulse is also related to average force F_{avg} and contact time Δt :

$$J = F_{\text{avg}} \Delta t \implies F_{\text{avg}} = \frac{|J|}{\Delta t}$$

$$F_{\text{avg}} = \frac{42}{0.020} = 2100 \text{ N}$$

Answer 1

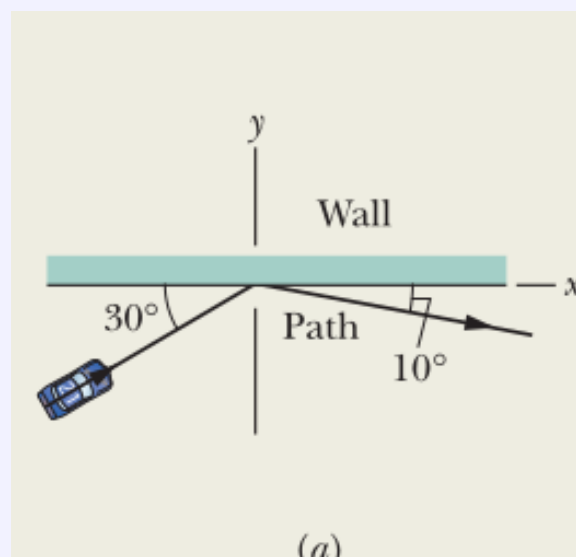
- (a) Impulse on the ball: $42 \text{ kg} \cdot \text{m/s}$ (upward)
- (b) Average force on the floor: 2100 N

Problem 2

Problem 2: Statement

Figure given below is an overhead view of the path taken by a race car driver as his car collides with the racetrack wall. Just before the collision, he is traveling at speed $v_i = 70$ m/s along a straight line at 30° from the wall. Just after the collision, he is traveling at speed $v_f = 50$ m/s along a straight line at -10° from the wall. His mass is $m = 80$ kg.

- (a) What is the impulse $\vec{J}$ on the driver due to the collision?
- (b) The collision lasts for 14 ms. What is the magnitude of the average force on the driver during the collision?



(a) Impulse on the Driver

From the impulse–momentum theorem:

$$\vec{J} = \vec{p}_f - \vec{p}_i = m(\vec{v}_f - \vec{v}_i).$$

We will evaluate the components separately.

X–Component of Impulse

$$\begin{aligned} J_x &= m(v_{fx} - v_{ix}) \\ &= (80 \text{ kg})[(50 \text{ m/s}) \cos(-10^\circ) - (70 \text{ m/s}) \cos(30^\circ)] \\ &= (80 \text{ kg})[50(0.9848) - 70(0.8660)] \\ &= (80 \text{ kg})[49.24 - 60.62] \\ &= (80 \text{ kg})(-11.38) \\ &= -910 \text{ kg} \cdot \text{m/s}. \end{aligned}$$

Y–Component of Impulse

$$\begin{aligned} J_y &= m(v_{fy} - v_{iy}) \\ &= (80 \text{ kg})[(50 \text{ m/s}) \sin(-10^\circ) - (70 \text{ m/s}) \sin(30^\circ)] \\ &= (80 \text{ kg})[(50)(-0.1736) - 70(0.5)] \\ &= (80 \text{ kg})[-8.68 - 35] \\ &= (80 \text{ kg})(-43.68) \\ &= -3495 \text{ kg} \cdot \text{m/s} \approx -3500 \text{ kg} \cdot \text{m/s}. \end{aligned}$$

Magnitude of Impulse

$$\begin{aligned} J &= \sqrt{J_x^2 + J_y^2} \\ &= \sqrt{(-910)^2 + (-3500)^2} \\ &= \sqrt{828,100 + 12,250,000} \\ &= \sqrt{13,078,100} \\ &= 3617 \text{ kg} \cdot \text{m/s}. \end{aligned}$$

Direction of Impulse

$$\theta = \tan^{-1}\left(\frac{J_y}{J_x}\right) = \tan^{-1}\left(\frac{-3500}{-910}\right) \approx 75.3^\circ$$

above the x -axis (toward the wall).

$$\boxed{\vec{J} = 3617 \text{ kg} \cdot \text{m/s at } 75.3^\circ.}$$

(b) Average Force During Collision

From the impulse–momentum relation:

$$\vec{J} = \vec{F}_{\text{avg}} \Delta t.$$

Hence,

$$F_{\text{avg}} = \frac{J}{\Delta t}.$$

Substituting the values:

$$F_{\text{avg}} = \frac{3617 \text{ kg} \cdot \text{m/s}}{0.014 \text{ s}} = 2.583 \times 10^5 \text{ N}.$$

$$F_{\text{avg}} \approx 2.6 \times 10^5 \text{ N}.$$

Interpretation: The driver experiences an enormous average force of about $2.6 \times 10^5 \text{ N}$ during the short 14-ms collision. This strong impulse drastically changes his velocity direction and magnitude.

Problem 3

Problem : Sliding Crate

A stevedore slides a crate along a dock with a 50 kg horizontal force of 175 N. The opposing force of friction is 120 N. If started from rest, what is the crate's final velocity after 0.5 s?

Solution 2

Net force :

$$F_{\text{net}} = F_{\text{applied}} - F_{\text{friction}} = 175 - 120 = 55 \text{ N}$$

Impulse : Impulse is the product of net force and contact time:

$$J = F_{\text{net}}\Delta t = 55 \times 0.5 = 27.5 \text{ N} \cdot \text{s}$$

Final velocity : Impulse equals change in momentum:

$$J = m(v_f - v_0) \implies 27.5 = 50(v_f - 0)$$

$$v_f = \frac{27.5}{50} = 0.55 \text{ m/s}$$

Answer 2

Final velocity of the crate: $v_f = 0.55 \text{ m/s}$

Problem 4

Problem : Force at an Angle

A 3 kg sled is moving on a frictionless ice surface with a velocity of 5 m/s to the right. A skier applies a force of 12 N at an angle of 60° above the horizontal for 2 s.

Questions: 1. Calculate the impulse delivered in the horizontal direction. 2. Find the horizontal velocity of the sled after the force is applied.

Solution 4

Horizontal component of force

$$F_x = F \cos \theta = 12 \cos 60^\circ = 12 \times 0.5 = 6 \text{ N}$$

Horizontal impulse

$$J_x = F_x \Delta t = 6 \times 2 = 12 \text{ N} \cdot \text{s}$$

Final horizontal velocity

$$J_x = m(v_{fx} - v_i) \implies 12 = 3(v_{fx} - 5)$$

$$v_{fx} - 5 = 4 \implies v_{fx} = 9 \text{ m/s}$$

Answer 4

Impulse in horizontal direction: $12 \text{ N} \cdot \text{s}$

Final horizontal velocity: 9 m/s

Problem 5

Problem : Variable Force

A 2 kg cart is initially at rest. A time-dependent horizontal force $F(t) = 4t$ N acts on the cart for 3 s.

Questions: 1. Find the total impulse delivered to the cart. 2. Determine the final velocity of the cart.

Solution 5

Total impulse Impulse is the integral of force over time:

$$J = \int_0^3 F(t) dt = \int_0^3 4t dt$$

$$J = 4 \left[\frac{t^2}{2} \right]_0^3 = 2(9 - 0) = 18 \text{ N} \cdot \text{s}$$

Final velocity

$$J = mv_f \implies 18 = 2v_f \implies v_f = 9 \text{ m/s}$$

Answer 5

Total impulse: $18 \text{ N} \cdot \text{s}$

Final velocity of the cart: 9 m/s

Problem 6

Problem : Impulse with Tension, Friction, and Variable Force

A 5 kg crate is on a horizontal surface with a kinetic friction force of 20 N. It is pulled by a rope with a time-dependent tension

$$T(t) = 50 + 10t \text{ N},$$

where t is in seconds. The crate starts from rest.

- (a) Determine the net impulse on the crate over the first 2 s.
- (b) Calculate the final velocity of the crate after 2 s.

Solution 6

Net force as a function of time

$$F_{\text{net}}(t) = T(t) - f_{\text{friction}} = (50 + 10t) - 20 = 30 + 10t \text{ N}$$

Impulse over 2 seconds

Impulse is the integral of net force over time:

$$J = \int_0^2 F_{\text{net}}(t) dt = \int_0^2 (30 + 10t) dt$$

$$J = \left[30t + 5t^2 \right]_0^2 = 30(2) + 5(4) = 60 + 20 = 80 \text{ N} \cdot \text{s}$$

Final velocity

$$J = m(v_f - v_0) \implies 80 = 5(v_f - 0)$$

$$v_f = \frac{80}{5} = 16 \text{ m/s}$$

Answer 6

- (a) Net impulse on the crate: 80 N · s
- (b) Final velocity of the crate: 16 m/s

Law of Conservation of Linear Momentum

Conservation of Linear Momentum

If the net external force $\vec{F}_{\text{net}}$ acting on a system of particles is zero, and the system is closed (no particles enter or leave), then from Newton's second law:

$$\frac{d\vec{P}}{dt} = 0$$

This implies that the total linear momentum $\vec{P}$ remains constant:

$$\vec{P} = \text{constant}$$

Law of Conservation of Linear Momentum:

If no net external force acts on a system of particles, the total linear momentum of the system cannot change.

For problem-solving, we express it as:

$$\vec{P}_i = \vec{P}_f$$

where $\vec{P}_i$ and $\vec{P}_f$ are the total momentum of the system at initial and later times, respectively.

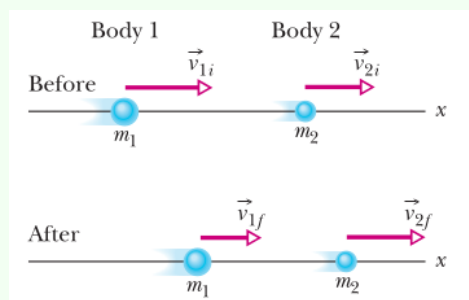
Conservation of Linear Momentum

- In a **closed, isolated system**, the total linear momentum $\vec{P}$ is conserved:

$$\vec{P}_i = \vec{P}_f.$$

In one dimension, this becomes:

$$m_1 v_{1i} + m_2 v_{2i} = m_1 v_{1f} + m_2 v_{2f}.$$



- The momentum conservation law applies to all collisions — elastic or inelastic.

Kinetic Energy in Collisions

Key Points

- The total kinetic energy of a system may or may not be conserved in a collision.

- **Elastic Collision:** Kinetic energy is conserved.

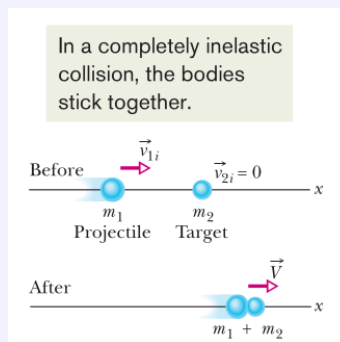
$$K_i = K_f.$$

$$\frac{1}{2}m_1v_{1i}^2 + \frac{1}{2}m_2v_{2i}^2 = \frac{1}{2}m_1v_{1f}^2 + \frac{1}{2}m_2v_{2f}^2$$

- **Inelastic Collision:** Some kinetic energy is converted into other forms (heat, sound, deformation).

$$\frac{1}{2}m_1v_{1i}^2 + \frac{1}{2}m_2v_{2i}^2 \neq \frac{1}{2}m_1v_{1f}^2 + \frac{1}{2}m_2v_{2f}^2$$

- **Completely Inelastic Collision:** The two objects stick together after the collision, resulting in maximum kinetic energy loss.



$$\frac{1}{2}m_1v_{1i}^2 + \frac{1}{2}m_2v_{2i}^2 \neq \frac{1}{2}(m_1 + m_2)V^2$$

$$m_1v_{1i} + m_2v_{2i} = (m_1 + m_2)V$$

- Everyday collisions (e.g., cars, baseball and bat) are typically inelastic.
- In a completely inelastic collision, momentum is conserved, but kinetic energy is not.

Collision in two dimension:

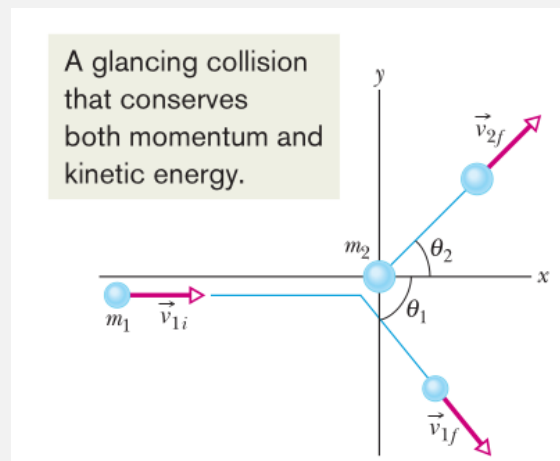


Figure: An elastic collision between two bodies where the collision is not head-on. The body with mass m_2 (the target) is initially at rest.

In a two-dimensional **elastic collision**, both **momentum** and **kinetic energy** are conserved.

The **law of conservation of linear momentum** is given by:

$$\vec{P}_{1i} + \vec{P}_{2i} = \vec{P}_{1f} + \vec{P}_{2f}.$$

This can be expressed in terms of components along the x - and y -axes as:

Along the x -axis:

$$m_1 v_{1i} = m_1 v_{1f} \cos \theta_1 + m_2 v_{2f} \cos \theta_2$$

Along the y -axis:

$$0 = m_1 v_{1f} \sin \theta_1 - m_2 v_{2f} \sin \theta_2$$

For an **elastic collision**, the total kinetic energy is also conserved:

$$\frac{1}{2} m_1 v_{1i}^2 + \frac{1}{2} m_2 v_{2i}^2 = \frac{1}{2} m_1 v_{1f}^2 + \frac{1}{2} m_2 v_{2f}^2$$

Problem 7:

Problem Statement:

A 1.0 kg ball is moving at 2.0 m s^{-1} to the right (along the x-axis). It collides with a stationary 1.0 kg ball. After the collision, the first ball moves at 1.0 m s^{-1} at an angle of 60° above the x-axis.

Assuming the collision is perfectly elastic and happens on a frictionless surface, find:

1. The speed of the second ball after the collision.
2. The direction of the second ball (angle below the x-axis).

Solution:

Given that:

- $m_1 = m_2 = 1.0 \text{ kg}$
- Initial velocity of ball 1: $v_{1i} = 2.0 \text{ m s}^{-1}$
- Final velocity of ball 1: $v_{1f} = 1.0 \text{ m s}^{-1}$ at $\theta_1 = 60^\circ$
- Final velocity of ball 2: unknown v_{2f}, θ_2

Conservation of Momentum in the x-direction

$$m_1 v_{1i} = m_1 v_{1f} \cos \theta_1 + m_2 v_{2f} \cos \theta_2 \quad (1)$$

$$1 \cdot 2.0 = 1 \cdot 1.0 \cdot \cos 60^\circ + 1 \cdot v_{2f} \cos \theta_2 \quad (2)$$

$$2.0 = 0.5 + v_{2f} \cos \theta_2 \quad (3)$$

$$v_{2f} \cos \theta_2 = 1.5 \quad (1)$$

Conservation of Momentum in the y-direction

$$0 = m_1 v_{1f} \sin \theta_1 - m_2 v_{2f} \sin \theta_2 \quad (4)$$

$$0 = 1.0 \cdot \sin 60^\circ - v_{2f} \sin \theta_2 \quad (5)$$

$$v_{2f} \sin \theta_2 = 0.866 \quad (2)$$

Dividing Equation (2) by Equation (1)

$$\tan \theta_2 = \frac{v_{2f} \sin \theta_2}{v_{2f} \cos \theta_2} = \frac{0.866}{1.5} = 0.577 \quad (6)$$

$$\theta_2 = \tan^{-1}(0.577) = 30^\circ \quad (7)$$

Solving for v_{2f} using Equation (2)

$$\sin \theta_2 = \frac{0.866}{v_{2f}} \Rightarrow \sin 30^\circ = \frac{0.5}{1} \quad (8)$$

$$0.5 = \frac{0.866}{v_{2f}} \quad (9)$$

$$v_{2f} = \frac{0.866}{0.5} = 1.732 \text{ m/s} \quad (10)$$

Answer :

- Speed of the second ball after collision: $\boxed{1.73 \text{ m/s}}$
- Direction of the second ball: $\boxed{30^\circ}$ below the x-axis

Problem 8:

Two bodies, A and B, each with a mass of 2.0 kg, collide. The velocities before the collision are:

$$\mathbf{v}_A = (15\hat{i} + 30\hat{j}) \text{ m/s}, \quad \mathbf{v}_B = (-10\hat{i} + 5\hat{j}) \text{ m/s}$$

After the collision, the velocity of body A is given by:

$$\mathbf{v}'_A = (-5\hat{i} + 20\hat{j}) \text{ m/s}$$

We are asked to find:

- (a) The final velocity of body B ($\mathbf{v}'_B$).
- (b) The change in the total kinetic energy of the system (including sign).

Solution

(a) Final velocity of body B

From the principle of conservation of linear momentum:

$$m_A \mathbf{v}_A + m_B \mathbf{v}_B = m_A \mathbf{v}'_A + m_B \mathbf{v}'_B$$

Since $m_A = m_B = m = 2.0 \text{ kg}$, the masses cancel out, and the equation becomes:

$$\mathbf{v}_A + \mathbf{v}_B = \mathbf{v}'_A + \mathbf{v}'_B$$

Substituting the given values:

$$(15\hat{i} + 30\hat{j}) + (-10\hat{i} + 5\hat{j}) = (-5\hat{i} + 20\hat{j}) + \mathbf{v}'_B$$

Simplifying:

$$(5\hat{i} + 35\hat{j}) = (-5\hat{i} + 20\hat{j}) + \mathbf{v}'_B$$

Now, solve for $\mathbf{v}'_B$:

$$\mathbf{v}'_B = (5\hat{i} + 35\hat{j}) - (-5\hat{i} + 20\hat{j})$$

$$\mathbf{v}'_B = (10\hat{i} + 15\hat{j}) \text{ m/s}$$

Thus, the final velocity of body B is:

$$\mathbf{v}'_B = (10\hat{i} + 15\hat{j}) \text{ m/s}$$

(b) Change in Total Kinetic Energy

The total kinetic energy before and after the collision is given by:

$$K_{\text{initial}} = \frac{1}{2}m_A v_A^2 + \frac{1}{2}m_B v_B^2$$
$$K_{\text{final}} = \frac{1}{2}m_A v_A'^2 + \frac{1}{2}m_B v_B'^2$$

The change in the total kinetic energy is:

$$\Delta K = K_{\text{final}} - K_{\text{initial}}$$

Initial Kinetic Energy:

$$K_{\text{initial}} = \frac{1}{2}(2.0) (15^2 + 30^2) + \frac{1}{2}(2.0) ((-10)^2 + 5^2)$$

$$K_{\text{initial}} = (1.0) (225 + 900) + (1.0) (100 + 25)$$

$$K_{\text{initial}} = (1.0) \times 1125 + 125 = 1250 \text{ J}$$

Final Kinetic Energy:

$$K_{\text{final}} = \frac{1}{2}(2.0) ((-5)^2 + 20^2) + \frac{1}{2}(2.0) (10^2 + 15^2)$$

$$K_{\text{final}} = (1.0) (25 + 400) + (1.0) (100 + 225)$$

$$K_{\text{final}} = (1.0) \times 425 + 325 = 750 \text{ J}$$

Change in Kinetic Energy:

$$\Delta K = 750 \text{ J} - 1250 \text{ J} = -500 \text{ J}$$

Thus, the change in kinetic energy is:

$$\Delta K = -500 \text{ J}$$